Condensed Novelty Check: Delay–Layout Coupling in UWB Self-Calibration

Grounded in the Vicon-validated Erlangen campaign.

1 Thesis

UWB anchor self-calibration couples antenna-delay/range-bias parameters with layout scale, so same-environment tag accuracy is not a reliable proxy for physical calibration quality. In the Erlangen campaign the production AutoPos self-calibrated layout (`v4-io`) is scale-biased (metric-incorrect): an under-absorbed common-mode range bias expands it to an AutoPos-to-Vicon Sim(3) scale of 0.958 (a range-bias signature, not a true geometric distortion), while a common-mode delay re-parameterization recovers metric scale (1.010) and nearly halves the rigid anchor RMSE (105.4 to 63.0 mm). Yet the metric-correct common-mode layout gives a *worse* same-environment static median (109.5 mm) than the scale-biased `v4-io` layout (72.7 mm) when the tag delay is left uncorrected, because the layout's scale bias compensates an uncorrected positive tag-device delay. Per-frame joint tag-delay estimation restores the common-mode layout to 72.9 mm median but destabilizes the tail (P95 317.5 mm). A scale-biased, metric-incorrect layout can therefore win in-environment *when the tag delay is left uncorrected*—once it is corrected with a single global constant the metric-correct layout becomes the better positioner (68.5 mm held-out median vs 72.7 mm; better at 17 of 24 positions, paired $p \approx 0.02$), so this is a conditional failure mode, not a claim that a metric-incorrect layout is inherently a better positioner. Mechanistically, the self-calibrated layout is internally self-consistent but not metric: the common-mode delay and isotropic layout scale span a near-degenerate direction ($\rho = -0.977$), so a family of (scale, delay) solutions fits the inter-anchor ranges equally well and range data alone cannot select the metrically true one. It is direct evidence that metric correctness and positioning accuracy must be validated separately.

2 Literature Gap

What is known	What is NOT known (our gap)
UWB self-calibration estimates anchors from radio ranges, robot motion, multihop links, or auxiliary sensing (Hamer & D'Andrea, 2018; Corbalan et al., 2023; Van Herbruggen et al., 2023; Almansa et al., 2020; Ledergerber et al., 2015; Ridolfi et al., 2021; Shi et al., 2019; Piavanini et al., 2022; Queralta et al., 2020).	These papers do not isolate whether the layout with the best tag accuracy is metrically correct, nor whether tag accuracy can rank a scale-biased layout above a validated metric one.
Antenna-delay and range-bias calibration are established UWB problems (Ledergerber & D'Andrea, 2018; De Preter et al., 2019; Shalaby et al., 2023; Liu et al., 2024; Shah et al., 2021, 2022).	In UWB ranging, delay/range-bias is usually a nuisance correction (in acoustic/TOA self-calibration it is a first-class jointly-solved unknown); its ability to absorb global layout-scale error in-environment is not characterized.
NLOS mitigation uses channel statistics, learned models, robust weighting, or biased-range models (Wymeersch et al., 2012; Di Franco et al., 2017; Marano et al., 2010; Guvenc et al., 2007; Venkatesh & Buehrer, 2007; Meghani et al., 2019; Jeong et al., 2023; Kong, 2023; Sung et al., 2023; Yang et al., 2024).	Structured positive range bias is not connected to which self-calibration geometry is physically valid.
Fisher/CRLB, rigidity, and observability analyses describe weak directions and rank deficiencies in localization and calibration networks (Shen et al., 2010; Aspnes et al., 2006; Patwari et al., 2005; Wymeersch et al., 2009; Prorok, 2013; Jourdan et al., 2008; Hesch et al., 2014; Goudar et al., 2021).	Prior work does not connect a coupled delay–scale direction to a measured scale-bias cancellation result (a scale-biased layout winning in-environment) in UWB self-calibration.
Scale–propagation-parameter ambiguity is known in acoustic self-calibration and related ranging systems (Burgess et al., 2015). AniTrack reported self-localized anchors outperforming surveyed anchors in tag accuracy (13.96 vs 16.57 cm, $n = 7$) (Luder et al., 2025).	The interaction with structured UWB range bias, and the resulting in-environment advantage of the scale-biased layout, is not reported. A consistent external observation (within one SD, competing explanations not excluded), not a diagnosis: no Sim(3) scale, delay–layout coupling, or causal intervention.

3 Our Contribution

The first contribution is scientific: we quantitatively characterize delay–layout coupling in UWB anchor self-calibration. A four-way ablation (self-calibrated vs Vicon-measured anchor coordinates $\times$ residual-correction treatments) shows the fitted residual corrections are frame-conditioned: Vicon-measured anchor coordinates degrade tag localization (311 mm RMSE with no correction, 252 mm with transplanted corrections) unless the corrections are re-estimated in the Vicon frame (78 mm). A common-mode delay re-parameterization ($d_i = c + e_i$, $c = 112$ mm) recovers metric scale from inter-anchor ranges alone (Sim(3) 0.958 $\to$ 1.010; rigid RMSE 105.4 $\to$ 63.0 mm); clamping the common mode to zero raises the solver cost by about 70%, and five perturbed initializations converge to the same scale. A gauge-free Fisher analysis on the real geometry confirms this is a strong-but-not-strict near-degeneracy: the common-mode delay correlates with the isotropic layout-scale mode at $\rho = -0.977$ (variance inflation 22$\times$; profiling the layout out of the delay flattens its likelihood 22$\times$; 1 mm common delay $\leftrightarrow$ -1.22 mm scale at the array edge; read from the delay direction's marginal, not the globally-softest raw eigenvector, which is a within-layer wiggle artifact). The decisive observation is that the metric-correct layout is the worse in-environment positioner *only* until the tag delay is corrected. The fair-comparison control points the same way: because the deployed firmware applies one generic antenna-delay default (16436) to every node with no per-device calibration, the physically honest correction is a single global tag-delay constant. With a leave-one-out, no-peek constant ($\approx$50 mm) the metric-correct layout becomes the better positioner—68.5 mm held-out median (P95 156 mm) vs the scale-biased 72.7 mm, better at 17 of 24 positions (paired $p \approx 0.02$; the marginal-median gap alone is within noise)—and with a far more stable tail than per-frame joint estimation (which only ties the median at 72.9 mm by overfitting, P95 317.5 mm). The scale-biased layout's apparent advantage was thus a correctable constant its scale error absorbed, not better physical calibration. An orthogonal hardware check confirms that the DWM1001C (DW1000) modules used here ship no usable per-device antenna-delay calibration: reading the factory OTP of all nine modules (the lone outlier verified by five repeated reads) shows the stored antenna delay is near-uniform (eight units 16472, one 16451) and does *not* track the fitted per-device delays—seven modules share one OTP value yet their fitted delays span ≈ 49–148 mm—while the per-device antenna-calibration temperature is unprogrammed throughout, even though the same OTP *does* carry per-device crystal-trim and temperature-sensor cal. Applying the OTP value is a global common-mode shift that cannot supply the per-device differences driving the coupling, so the coupling is intrinsic to these commodity single-antenna modules, not a skipped calibration step.

Separately, the same campaign exhibits a *tag-side* face of the coupling: once the balanced two-layer array brings the vertical into its well-conditioned regime, a common tag range bias aliases predominantly into height (VDOP/HDOP ≈ 1.6), and an active vertical-injection control produced no dose-response. This is the residual-vertical limit reported by the companion system paper.

The second contribution is methodological: a Vicon-grounded falsification protocol that separates metric correctness from same-environment positioning accuracy. It uses Sim(3) anchor-scale validation, a layout/delay transfer ablation, per-anchor range-bias decomposition with error-budget closure, phase-center sensitivity, and nested spatial CV, so that a single accuracy number cannot hide the mechanism.

As a discussion-level conjecture (not a contribution), a similar failure mode could occur wherever per-device bias and geometric scale share a weakly observable direction (GNSS clock–geometry, acoustic sound-speed–geometry); the analogy is imperfect because delay is additive while scale is multiplicative, so cancellation is only approximate over the working-distance range.

A raw-frame intervention that reduces the positive tag-anchor range tail reverses the ranking in favour of the metric-correct layout; this causal intervention is the central cancellation evidence.

Priority and disclosure. We claim not the bare existence of the coupling (acoustic/GNSS/AniTrack precedents preclude that) but: (i) the first gauge-free Fisher quantification of the delay↔scale near-degeneracy on real UWB hardware ($\rho = -0.977$, 22× variance inflation, 1 mm delay ↔ −1.22 mm edge-scale); (ii) the conditional wrong-geometry-wins regime and the control that closes it (a single global tag-delay constant makes the metric-correct layout the better positioner, 68.5 vs 72.7 mm); and (iii) the Vicon-grounded falsification protocol. For defensive-publication value the enabling three-condition combination is disclosed once (range-only self-calibration + independent metric ground truth for the geometry + multi-parameterization comparison), and the Fisher analysis ships as a runnable artifact so the numbers are independently reproducible. Citable: *"the common-mode antenna delay is weakly observable against isotropic layout scale ($\rho = -0.977$ on the deployed array), so same-environment accuracy can rank a scale-biased layout above a metric-correct one until a single global delay constant is applied—after which the metric-correct layout becomes the better positioner."*

4 Nearest Prior Art

Paper	What they did	How we differ
Hamer & D'Andrea (2018)	Built a self-calibrating one-way UWB network with clock synchronization and anchor computation for multi-robot localization.	We test whether the empirically best radio-only layout is physically correct, and show that a scale-biased layout can win in-environment (under an uncorrected tag delay) by cancelling structured range bias.
Lutz et al. (2019)	Used visual-inertial SLAM and fiducial markers to estimate UWB anchor poses and sensor error-model parameters.	Their calibration is aided by external visual structure; ours diagnoses a range-only self-calibration degeneracy using Vicon only as independent validation.
Ledergerber & D'Andrea (2018)	Estimated module-dependent range inaccuracies by maximum likelihood without external ground truth.	We show that fitting delay/range inaccuracies can reward a metric-incorrect (scale-biased) solution unless anchor metric scale is checked separately.
Luder et al. (2025) (AniTrack)	Reported that self-localized DWM3000 anchors gave better tag accuracy than laser-surveyed anchors (13.96 vs 16.57 cm average error).	They report an unexplained, small-sample paradox; we propose and Vicon-test a candidate mechanism via Sim(3) scale, a four-way layout/delay ablation, and common-mode scale recovery.
Pan et al. (2015) (GNSS)	Showed satellite-clock estimation can absorb orbit error and improve PPP positioning accuracy.	Their coupled-error absorption is a GNSS clock/orbit effect; our result is a UWB self-calibrated-layout scale bias that makes the metric-incorrect layout beat the correct one when the tag bias is uncorrected.

5 Risk and Status

Risk	Severity	One-line mitigation
One room and 24 static positions	High	State that transfer remains unproven and freeze the pipeline for second- and N-room validation.
Results presuppose full 8-anchor participation in each fix	Medium	All figures derive from 3D fixes using all eight anchors of the 4+4 array (sequential vs broadcast is irrelevant for a static fix); fixes using fewer anchors (e.g., a 4-anchor subset) hit a much larger vertical CRLB or, for coplanar subsets, the singular regime, and are not expected to reproduce them.
Cross-domain scale–delay ambiguity is prior art	Medium	Claim quantitative UWB characterization plus the in-environment cancellation result, not the bare ambiguity.
Common-mode scale recovery is a diagnostic	Medium	Present it as a mechanism re-parameterization, not as the deployed headline pipeline.
High-error tail remains	Medium	Separate median mechanism evidence from P95/dynamic performance claims.
Phase-center uncertainty	Low	Use sensitivity analysis and avoid treating the Vicon-oracle rank as the sole proof.

No prior work we found explicitly characterizes the in-environment delay–layout cancellation, though De Preter (2019), Piavanini (2022), Batstone (2017), Hamer & D'Andrea (2018) and the acoustic/GNSS work are close and are engaged directly. Target: *Measurement* (IF ∼5.6). Working title: "Metric Correctness versus Positioning Accuracy in UWB Anchor Self-Calibration: A Vicon-Grounded Falsification Study."

References

Almansa, C.M., Shule, W., Queralta, J.P. and Westerlund, T., 2020. Autocalibration of a mobile UWB localization system for ad-hoc multi-robot deployments in GNSS-denied environments. *arXiv preprint arXiv:2004.06762*.

Aspnes, J., Eren, T., Goldenberg, D.K., Morse, A.S., Whiteley, W., Yang, Y.R., Anderson, B.D.O. and Belhumeur, P.N., 2006. A theory of network localization. *IEEE Transactions on Mobile Computing*, 5(12), pp.1663–1678.

Burgess, S., Kuang, Y. and Åström, K., 2015. TOA sensor network self-calibration for receiver and transmitter spaces with difference in dimension. *Signal Processing*, 107, pp.33–42.

Corbalan, P., Picco, G.P., Coors, M. and Jain, V., 2023. Self-localization of ultra-wideband anchors: From theory to practice. *IEEE Access*, 11, pp.29711–29725.

De Preter, A., Goysens, G., Anthonis, J., Swevers, J. and Pipeleers, G., 2019. Range bias modeling and autocalibration of a UWB positioning system. In: *International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, pp.1–8.

Di Franco, C., Prorok, A., Atanasov, N., Kempke, B., Dutta, P., Kumar, V. and Pappas, G.J., 2017. Calibration-free network localization using non-line-of-sight UWB measurements. In: *ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*.

Goudar, A., Zhao, W. and Schoellig, A.P., 2021. Online spatio-temporal calibration of tightly-coupled ultrawideband-aided inertial localization. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Guvenc, I., Chong, C.-C., Watanabe, F. and Inamura, H., 2007. NLOS identification and weighted least-squares localization for UWB systems using multipath channel statistics. *EURASIP Journal on Advances in Signal Processing*, 2007.

Hamer, M. and D'Andrea, R., 2018. Self-calibrating ultra-wideband network supporting multi-robot localization. *IEEE Access*, 6, pp.22292–22304.

Van Herbruggen, B., Luchie, S., Fontaine, J. and De Poorter, E., 2023. Multihop self-calibration algorithm for ultra-wideband anchor node positioning. *IEEE Journal of Indoor and Seamless Positioning and Navigation*.

Hesch, J.A., Kottas, D.G., Bowman, S.L. and Roumeliotis, S.I., 2014. Camera-IMU-based localization: Observability analysis and consistency improvement. *The International Journal of Robotics Research*, 33(1), pp.182–201.

Jeong, S.-G., Do, Q.-V., Hwang, H.-J., Hasegawa, M., Sekiya, H. and Hwang, W.-J., 2023. Hybrid quantum convolutional neural networks for UWB signal classification. *IEEE Access*.

Jourdan, D.B., Dardari, D. and Win, M.Z., 2008. Position error bound for UWB localization in dense cluttered environments. *IEEE Transactions on Aerospace and Electronic Systems*, 44(2), pp.613–628.

Kong, Q., 2023. NLOS identification for UWB positioning based on IDBO and convolutional neural networks. *IEEE Access*.

Ledergerber, A., Hamer, M. and D'Andrea, R., 2015. A robot self-localization system using one-way ultra-wideband communication. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp.3131–3137.

Ledergerber, A. and D'Andrea, R., 2018. Calibrating away inaccuracies in ultra wideband range measurements: A maximum likelihood approach. *IEEE Access*, 6, pp.78719–78730.

Liu, Z., Hakala, T., Hyyppa, J., Kukko, A., Kaartinen, H. and Chen, R., 2024. Data-driven antenna delay calibration for UWB devices for network positioning. *IEEE Transactions on Instrumentation and Measurement*.

Lutz, P., Schuster, M.J. and Steidle, F., 2019. Visual-inertial SLAM aided estimation of anchor poses and sensor error model parameters of UWB radio modules. In: *International Conference on Advanced Robotics (ICAR)*, pp.739–746.

Luder, V., Schulthess, L., Cortesi, S., Rivero Davis, L. and Magno, M., 2025. AniTrack: A power-efficient, time-slotted and robust UWB localization system for animal tracking in a controlled setting. *arXiv preprint arXiv:2506.00216*.

Marano, S., Gifford, W.M., Wymeersch, H. and Win, M.Z., 2010. NLOS identification and mitigation for localization based on UWB experimental data. *IEEE Journal on Selected Areas in Communications*.

Meghani, S.K., Asif, M., Awin, F. and Tepe, K., 2019. Empirical based ranging error mitigation in IR-UWB: A fuzzy approach. *IEEE Access*.

Pan, S., Chen, W., Jin, X., Shi, X. and He, F., 2015. Real-time PPP based on the coupling estimation of clock bias and orbit error with broadcast ephemeris. *Sensors*, 15(7), pp.17808–17826.

Patwari, N., Ash, J.N., Kyperountas, S., Hero, A.O., Moses, R.L. and Correal, N.S., 2005. Locating the nodes: Cooperative localization in wireless sensor networks. *IEEE Signal Processing Magazine*, 22(4), pp.54–69.

Piavanini, M., Barbieri, L., Brambilla, M., Cerutti, M., Ercoli, S., Agili, A. and Nicoli, M., 2022. A calibration method for antenna delay estimation and anchor self-localization in UWB systems. In: *IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0&IoT)*, pp.173–177.

Prorok, A., 2013. *Models and Algorithms for Ultra-Wideband Localization in Single- and Multi-Robot Systems*. PhD thesis. EPFL.

Queralta, J.P., Almansa, C.M., Schiano, F., Floreano, D. and Westerlund, T., 2020. UWB-based system for UAV localization in GNSS-denied environments. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Ridolfi, M., Kaya, A., Berkvens, R., Weyn, M., Joseph, W. and De Poorter, E., 2021. Self-calibration and collaborative localization for UWB positioning systems. *ACM Computing Surveys*.

Shah, S., Chaiwong, K., Kovavisaruch, L., Kaemarungsi, K. and Demeechai, T., 2021. Antenna delay calibration of UWB nodes. *IEEE Access*, 9, pp.63294–63305.

Shah, S., Kovavisaruch, L., Kaemarungsi, K. and Demeechai, T., 2022. Node calibration in UWB-based RTLSs using multiple simultaneous ranging. *Sensors*, 22(3), 864.

Shalaby, M., Cossette, C.C., Forbes, J.R. and Le Ny, J., 2023. Calibration and uncertainty characterization for ultra-wideband two-way-ranging measurements. In: *IEEE International Conference on Robotics and Automation (ICRA)*.

Shen, Y., Wymeersch, H. and Win, M.Z., 2010. Fundamental limits of wideband localization—Part II: Cooperative networks. *IEEE Transactions on Information Theory*, 56(10), pp.4981–5000.

Shi, Q., Zhao, S., Cui, X., Lu, M. and Jia, M., 2019. Anchor self-localization algorithm based on UWB ranging and inertial measurements. *Tsinghua Science and Technology.*

Sung, S., Kim, H. and Jung, J.-i., 2023. Accurate indoor positioning for UWB-based personal devices using deep learning. *IEEE Access.*

Venkatesh, S. and Buehrer, R.M., 2007. Non-line-of-sight identification in ultra-wideband systems based on received signal statistics. *IET Microwaves, Antennas & Propagation.*

Wymeersch, H., Lien, J. and Win, M.Z., 2009. Cooperative localization in wireless networks. *Proceedings of the IEEE.*

Wymeersch, H., Marano, S., Gifford, W.M. and Win, M.Z., 2012. A machine learning approach to ranging error mitigation for UWB localization. *IEEE Transactions on Communications.*

Yang, H., Wang, Y., Seow, C., Sun, M. and Plets, D., 2024. UWB NLOS identification and mitigation based on bidirectional encoder representations from transformer (BERT) deep learning. In: *International Conference on Indoor Positioning and Indoor Navigation (IPIN).*